

Feature Selection in Statistical Machine Learning

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Lecture 2

Plan for today

- Re-visiting example
- Feature selection in a broader data science/machine learning context
- History and overview
- Feature selection definition

Recall: Housing data example

Y : House price; X_1 : Square footage; X_2 : Number of bedrooms; X_3 : Neighborhood quality score

- Correlation screening (filter method) suggests that all 3 features should be selected
- Multiple regression (embedded method) suggests that only X_1 and X_3 should be selected
- Why?

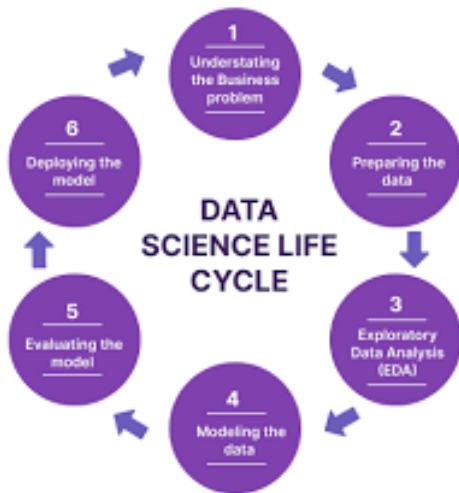
Regression vs correlation

- Correlation only considers the individual relationship between feature X_j and response Y
- Regression considers the relationship between X_j and Y *conditioned* on all other features
- In housing data example, once we have the square footage X_1 , knowing the number of bedrooms X_2 adds no further information
- X_1 and X_2 highly correlated and individually impact on house price Y but collectively, we only need X_1
- Feature selection gets really interesting and complex when there is dependence among features which is almost always the case
- In this case only 3 features. Imagine what may happen if there are hundreds, thousands or millions of features!

Marginal vs conditional independence/dependence

- Marginal dependence/independence focuses individual relationship between X_j and Y ignoring all other features (e.g. correlation, mutual information, filter methods, e.t.c.)
- Conditional dependence/independence focuses on the relationship between X_j and Y taking all the other features into account (e.g. regression, embedded and wrapper methods, e.t.c.)
- Focusing on this distinction is one of the most important aspects of this course and in probability/statistics more generally
- We will make this mathematically precise shortly (weeks 2-3)

Data science life cycle



Feature selection in the data science life cycle

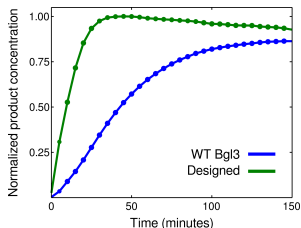
- (1) Begin with domain-specific context and dataset. What problem are you trying to solve and how does the data help you solve that?
- (2) Determine how the domain-specific context relates to feature selection (e.g. causality/intervention, decision-making, interpretability, scalability, e.t.c.).
- (3) Pre-process/clean response(s) and features so that they are in a form that makes the most sense relating to (1).
- (4) Understand data and prediction model to inform which feature selection algorithm(s) you want to use.
- (5) Validate the outputs of your feature selection algorithm (we discuss validation around weeks 7-9).
- (6) Connect back to (1) and (2) and determine conclusions and next steps.

Example 1: Housing data example

- (1) How should we set house price based on the 3 features of interest?
- (2) Which features are going to determine whether house price goes up or down?
- (3) Consider whether standardization or transforms for the features or responses are required.
- (4) Is linear model a good choice here (or would something that incorporates non-linearities be more appropriate)?
- (5) Validate the outputs of your feature selection algorithm (we discuss validation around weeks 7-9).
- (6) As discussed, based on regression results number of bedrooms is not an important feature based on the data if the square footage does not change. Model could consist of square footage and neighborhood quality only.

Example 2: Protein synthesis

- X data is protein sequences and Y is sugar/glucose production/yield
- Features are amino acid presences in different protein locations
 $p = 500^{20}$
- We developed a feature selection algorithm and selected 10 most significant features to intervene on



Example 2: Protein synthesis example

- (1) Can we engineer proteins or other biological systems to improve performance?
- (2) Determine which sequence positions should be intervened on and how should intervention occur.
- (3) Look for missing data and determine how best to deal with that.
- (4) Since responses are binary (functional or not) and feature dimensionality is extremely, some kind of regularized logistic regression may be suitable.
- (5) Can perform actual validation in the lab in this case.
- (6) Based on (5) it is clear that changing the protein structure significantly improves functionality.

Historical perspective and course overview

- 1930s (origins in Statistics)
- Regression subset selection (1950s)
- Pattern recognition/ML community comes in (1970s)
- Conditional independence perspective (1970s-80s)
- Feature Selection meets Machine Learning (1990s)
- Embedded Methods & Sparsity (2000s)
- High-Dimensional Statistics & causaling (2010s)
- Neural networks, LLMs and more (2020s)

1930s, origins in statistics

- Fisher's linear discriminant analysis (1936): effectively uses selected linear combinations of features for classification
- FS at this stage was ad hoc - removing *irrelevant* predictors to simplify regression (Week 6)

1950s, regression and subset selection

- Mallows's C_p , AIC (Akaike, 1974), and later BIC (Schwarz, 1978) provided criteria for model comparison (Week 6)
- Stepwise regression (forward, backward, stepwise selection) became standard in applied statistics (Week 7)
- Motivation: computational limits + interpretability

1970s, Pattern recognition/ML community comes in

- Pattern recognition community (e.g., Fukunaga's Introduction to Statistical Pattern Recognition, 1972) systematically studied FS for classification
- Early filter methods: mutual information, correlation-based feature ranking (Week 5)
- Wrapper methods: greedy search with error estimates on training/validation sets (Weeks 7, 8)

1970s-80s, Conditional independence perspective

- Dawid's formalization of conditional independence (1979) in statistics later influenced Markov blanket-based FS methods (Weeks 2,3, 10)
- Graphical models (Week 10)

1990s, Feature selection meets machine learning

- Explosion of interest due to text classification and microarray gene expression data (many features, few samples)
- Guyon & Elisseeff (2003) survey paper: canonical reference on FS in ML
- Formal categorization of filter methods, wrapper methods and embedded methods (Weeks 5,6,7)
- Computational focus: greedy algorithms, cross-validation, regularization

2000s, Embedded Methods and Sparsity

- Lasso (Tibshirani, 1996) became widely adopted in the 2000s as embedded FS via L1 regularization (Week 6)
- Support Vector Machines (SVMs) with recursive feature elimination (RFE) (Week 7)
- Decision trees and ensembles (Random Forests, 2001; Gradient Boosting) provided feature importance measures (Week 6, 10)
- FS became standard in bioinformatics, text mining, computer vision

2010s, High-dimensional statistics and causality

- Conditional independence testing + graphical models (Markov blankets, PC/FCI algorithms) connected FS to causal discovery (Week 2,3, 10)
- Nonlinear FS methods (kernel-based CI tests, HSIC, mRMR)
- Stability selection (Week 8)
- Deep learning sparked debate: do we still need FS when models 'learn representations'? (Week 12)

2020s, Neural networks, LLMs and more

- Interpretability and fairness → FS helps identify spurious predictors (Week 12)
- Automated ML (AutoML) → FS integrated into pipelines
- Massive data regimes → feature pruning for large language models and neural networks (Week 12)

Mathematical setup

Data $\{Y^{(i)}, X_1^{(i)}, X_2^{(i)}, \dots, X_p^{(i)}\}_{i=1}^n$

E.g. in housing data example $n = 300$, $p = 3$ (X 's and Y 's) are random variables

Statistical model:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = f(X_1, X_2, \dots, X_p)$$

Subset selection:

$$S^* \in \arg \min_{S \subseteq \{X_1, X_2, \dots, X_p\}, |S| \leq k} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

where $\mathcal{L}(., .)$ is a loss function (e.g. $\mathcal{L}(Y, f(X)) = (Y - f(X))^2$) and f is the prediction function to be learned in the function class $\mathcal{F}$

Note: S^* is not always unique

Unpacking some notation

$\mathbb{E}[\cdot]$ is *expected value* of a random variable and

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p]$$

refers to a *conditional expected value* of the response Y given features $(X_1, X_2, \dots, X_p)$. A simplification of

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = f(X_1, X_2, \dots, X_p)$$

that may be more interpretable is:

$$Y = f(X_1, X_2, \dots, X_p) + w$$

w is a noise random variable where $\mathbb{E}[w] = 0$.

Note that all prediction functions in supervised learning are conditional expectations at heart.

Question: When will the selected subset not be unique?

Example: Cholesterol and heart disease

	LDL	HDL	Total	Heart Disease
1	144.901425	63.8926632	208.794088	1
2	125.852071	78.6412496	204.493321	0
3	149.430656	29.0214864	178.452143	1
4	175.690896	58.4445386	234.135434	1
5	122.975399	40.2403615	163.21576	1
6	122.975891	42.6931192	165.669011	0
7	177.376385	41.1140911	218.490476	1
8	153.023042	37.0401385	190.06318	1
9	115.915768	50.7278244	166.643593	0
10	146.276801	37.5357483	183.81255	1

Question: When will the selected subset not be unique?

- Interchangeable/highly correlated features
- Example: Heart disease highly dependent on cholesterol ($n = 500$, $p = 3$)
- 3 different cholesterol measures: LDL (bad), HDL (good), Total = LDL + HDL
- Logistic regression selects all 3 features (and may be unstable at times)
- Logistic Lasso drops the 'Total' feature
- **Exercise:** Run both logistic regression and logistic Lasso and compare

Example: Housing data

Y : House price; X_1 : Square footage; X_2 : Number of bedrooms; X_3 : Neighborhood quality score

$$\mathbb{E}[Y|X_1, X_2, X_3] = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

or

$$Y = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + w$$

General feature selection:

$$S^* \in \arg \min_{S \subseteq \{1, 2, \dots, p\}, |S| \leq k} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

Feature selection for this example:

$$S^* \in \arg \min_{S \subseteq \{X_1, X_2, X_3\}, |S| \leq k} \min_{\beta_S} \mathbb{E}[(Y - \sum_{j \in S} X_j \beta_j)^2]$$

Note: Distinction between population and empirical quantities

General feature selection:

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Empirical (estimated quantities):

$$\widehat{S}^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq k} \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(Y^{(i)}, f(X_S^{(i)}))$$

Housing data example:

$$\widehat{S}^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq k} \min_{\beta_S} \frac{1}{n} \sum_{i=1}^n (Y^{(i)} - \sum_{j \in S} X_j^{(i)} \beta_j)^2$$